

1.1 Relations

1. A **relation** from set A to set B is defined as

(a) only a subset of A (b) a subset of $A \times B$ (the Cartesian product) (c) the union of A and B (d) any set containing both A and B

3. The **domain** of a relation R from A to B is the set of

(a) first elements of the ordered pairs of R (b) second elements of the ordered pairs of R (c) elements in B never used by R (d) all elements of $A \times B$

5. $R = \{(x, y) : y = x + 1, x \in \{1, 2, 3\}\}$. The range of R is

(a) $\{2, 3, 4\}$ (b) $\{1, 2, 3\}$ (c) $\{0, 1, 2\}$ (d) $\{1, 2, 3, 4\}$

7. A relation R is **symmetric** if

(a) $(x, x) \in R$ for all x (b) $(x, y) \in R$ implies $(y, x) \in R$ (c) $(x, y), (y, z) \in R$ imply $(x, z) \in R$ (d) $R = \emptyset$

1.2 & 1.3 Functions, Domain and Range

9. A **function** $f : A \rightarrow B$ is a relation in which

(a) every element of A has one image in B (b) some elements of A have two images (c) every element of B is an image (d) $A = B$

11. The domain of $f(x) = \sqrt{x-2}$ is

(a) $\{x : x \geq 2\}$ (b) $\{x : x > 2\}$ (c) $\mathbb{R}$ (d) $\{x : x \leq 2\}$

13. The domain of $f(x) = \frac{1}{x-3}$ is

(a) $\mathbb{R}$ (b) $\{x : x \neq 3\}$ (c) $\{x : x > 3\}$ (d) $\{x : x < 3\}$

15. If $f(x) = x^2 + 1$, then $f(-3) =$

(a) -8 (b) -10 (c) 10 (d) 7

17. If $f(x) = 4x - 3$ and $g(x) = 2x + 5$, then $(f + g)(x) =$

(a) $6x + 2$ (b) $6x - 2$ (c) $2x - 8$ (d) $6x + 8$

19. With f and g as above, $(fg)(x) =$

(a) $8x^2 + 14x - 15$ (b) $8x^2 - 14x + 15$ (c) $8x^2 + 26x - 15$ (d) $6x^2 + 2x$

2. If $A = \{2, 4, 6\}$, which ordered pair belongs to the relation $\leq \subset A \times A$?

(a) $(6, 4)$ (b) $(4, 6)$ (c) $(6, 2)$ (d) $(4, 2)$

4. For $R = \{(1, a), (2, b), (3, c)\}$, the range of R is

(a) $\{1, 2, 3\}$ (b) $\{a, b, c\}$ (c) $\{1, a\}$ (d) $\{a, b\}$

6. A relation R is **reflexive** on A if for all $x \in A$

(a) $(x, y) \in R$ implies $(y, x) \in R$ (b) $(x, x) \in R$ (c) $(x, x) \notin R$ (d) (x, y) and (y, z) in R imply $(x, z) \in R$

8. An **equivalence relation** is one that is

(a) reflexive and symmetric only (b) reflexive, symmetric and transitive (c) reflexive and transitive only (d) symmetric and transitive only

10. Which of the following relations R from A to B is a function?

(a) $\{(1, a), (1, b), (2, c)\}$ (b) $\{(1, a), (2, b), (3, c)\}$ (c) $\{(1, a), (2, a), (2, b)\}$ (d) $\{(1, a), (1, b), (1, c)\}$

12. The range of $f(x) = x^2$ is

(a) $\mathbb{R}$ (b) $\{y : y \geq 0\}$ (c) $\{y : y > 0\}$ (d) $\{y : y \leq 0\}$

14. If $f(x) = 3x - 2$, then $f(4) =$

(a) 12 (b) 10 (c) 14 (d) 6

16. The range of $f(x) = 2x + 1$ over domain $\{1, 2, 3\}$ is

(a) $\{3, 5, 7\}$ (b) $\{1, 2, 3\}$ (c) $\{0, 2, 4\}$ (d) $\{2, 4, 6\}$

18. With $f(x) = 4x - 3$ and $g(x) = 2x + 5$, $(f - g)(x) =$

(a) $2x - 8$ (b) $6x + 2$ (c) $2x + 8$ (d) $8x + 2$

20. $(f \circ g)(x)$ means

(a) $g(f(x))$ (b) $f(g(x))$ (c) $f(x) \cdot g(x)$ (d) $f(x) + g(x)$

21. If $f(x) = 2x$ and $g(x) = x + 3$, then $(f \circ g)(x) =$

- (a) $2x + 6$ (b) $2x + 3$ (c) $(x + 3)(2x)$ (d) $2(x^2) + 3$

22. If $f(x) = 2x$ and $g(x) = x + 3$, then $(g \circ f)(x) =$

- (a) $2x + 6$ (b) $2x + 3$ (c) $2x^2 + 6$ (d) $2 + 3x$

1.4 Types of Functions

23. A function $f : A \rightarrow B$ is **one-to-one** (injective) if

- (a) $f(a_1) = f(a_2)$ implies $a_1 = a_2$ (b) every element of B is an image (c) f is constant (d) both A and B are finite

25. A function that is both one-to-one and onto is called

- (a) a constant function (b) an identity function (c) a bijection (d) a partial function

27. $f(x) = x^2$ over $\mathbb{R}$ is an example of a function that is

- (a) one-to-one and onto (b) one-to-one only (c) onto only (d) neither one-to-one nor onto

29. An **identity function** is defined by

- (a) $f(x) = 0$ (b) $f(x) = x$ (c) $f(x) = 1$ (d) $f(x) = x^2$

24. A function $f : A \rightarrow B$ is **onto** (surjective) if

- (a) f is one-to-one (b) every element of B is the image of some element of A (c) no two elements share an image (d) f is injective and injective only

26. Which of the following is a constant function?

- (a) $f(x) = x$ (b) $f(x) = 5$ (c) $f(x) = 2x + 1$ (d) $f(x) = x^2$

28. $f(x) = 2x + 3$ over $\mathbb{R}$ is a function that is

- (a) one-to-one and onto (b) one-to-one but not onto (c) onto but not one-to-one (d) neither

1.5 Inverse of a Function

30. The inverse of $f(x) = 2x + 6$ is

- (a) $f^{-1}(x) = \frac{x-6}{2}$ (b) $f^{-1}(x) = \frac{x+6}{2}$ (c) $f^{-1}(x) = 2x - 6$ (d) $f^{-1}(x) = -2x - 6$

32. A function f has an inverse if and only if f is

- (a) onto (b) one-to-one (bijective for $f : A \rightarrow A$ of finite sets) (c) constant (d) even

34. The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are symmetric about the line

- (a) $y = 0$ (b) $x = 0$ (c) $y = x$ (d) $y = -x$

36. The inverse of $f(x) = x^3$ is

- (a) $f^{-1}(x) = \sqrt[3]{x}$ (b) $f^{-1}(x) = x^{\frac{1}{2}}$ (c) $f^{-1}(x) = \frac{1}{x^3}$ (d) $f^{-1}(x) = x^2$

31. The inverse of $f(x) = \frac{x}{3} - 4$ is

- (a) $f^{-1}(x) = 3x + 12$ (b) $f^{-1}(x) = 3x - 12$ (c) $f^{-1}(x) = \frac{x+4}{3}$ (d) $f^{-1}(x) = \frac{x}{3} + 4$

33. If $f(3) = 7$, then $f^{-1}(7) =$

- (a) 7 (b) 3 (c) -3 (d) 21

35. If $f(x) = 3x - 9$ and $f^{-1}(x) = \frac{x+9}{3}$, then $(f \circ f^{-1})(5) =$

- (a) 0 (b) 5 (c) 15 (d) 24

1.6 Absolute Value, Signum and Piecewise Functions

37. The signum function $\text{sgn}(x) = \frac{x}{|x|}$ for $x \neq 0$. Then $\text{sgn}(-7) =$

- (a) 0 (b) 1 (c) -1 (d) 7

39. Let $f(x) = \begin{cases} x+1 & \text{if } x < 2 \\ 3 & \text{if } x \geq 2 \end{cases}$. Then $f(1) =$

38. If $f(x) = |x|$, then $f(-4) =$

- (a) 4 (b) -4 (c) 16 (d) 0

40. For the same piecewise function f as above, $f(5) =$

(a) 3 (b) 1 (c) 4 (d) 2

41. The signum function $\operatorname{sgn}(x)$ takes the value 0 exactly when

(a) $x > 0$ (b) $x < 0$ (c) $x = 0$ (d) x is any real number

1.7 Even and Odd Functions

42. Which of the following is an **even** function over $\mathbb{R}$?

(a) $f(x) = x^3$ (b) $f(x) = x^4$ (c) $f(x) = 2x$ (d) $f(x) = x + 1$

1.8 Composition of Functions

44. If $f(x) = x + 1$ and $g(x) = 2x$, then $(f \circ g)(3) =$

(a) 6 (b) 7 (c) 8 (d) 9

46. If f and g are inverse functions, then $(f \circ g)(x) =$

(a) 0 (b) x (c) $f(x)$ (d) 1

1.9 Domain, Range and the Line Tests

48. The domain of $f(x) = \sqrt{x^2 - 9}$ is

(a) $\{x : -3 \leq x \leq 3\}$ (b) $\{x : x \leq -3 \text{ or } x \geq 3\}$
(c) $\{x : x \geq 3\}$ (d) $\{x : x \leq -3\}$

50. The vertical line test is used to decide whether a relation is

(a) one-to-one (b) onto (c) a function (d) invertible

1.10 More on Inverse Functions

52. The inverse of $f(x) = 5 - 3x$ is

(a) $f^{-1}(x) = \frac{5+x}{3}$ (b) $f^{-1}(x) = \frac{5-x}{3}$ (c) $f^{-1}(x) = \frac{x-5}{3}$ (d) $f^{-1}(x) = 3x - 5$

54. For $f(x) = x^2$ with domain $\{x : x \geq 0\}$, the inverse is

(a) $f^{-1}(x) = \sqrt{x}$ (b) $f^{-1}(x) = -\sqrt{x}$ (c) $f^{-1}(x) = x^2$ (d) $f^{-1}(x) = \frac{1}{x}$

56. The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of each other across the line

(a) $y = x$ (b) the x -axis (c) the y -axis (d) the origin

58. Which function has the same graph as its own inverse?

(a) 3 (b) 6 (c) 5 (d) 2

43. Which of the following is an **odd** function over $\mathbb{R}$?

(a) $f(x) = x^2 - 1$ (b) $f(x) = |x|$ (c) $f(x) = x^3$ (d) $f(x) = x^2$

45. If $f(x) = x^2$ and $g(x) = x - 1$, then $(g \circ f)(2) =$

(a) 3 (b) 5 (c) 9 (d) -3

47. If $f(x) = 3x$ and $g(x) = \frac{x}{3}$, then $(g \circ f)(x) =$

(a) $\frac{x}{9}$ (b) $9x$ (c) x (d) x^2

49. The range of $f(x) = \frac{1}{x}$ for $x \neq 0$ is

(a) $\{y : y \neq 0\}$ (b) $\{y : y > 0\}$ (c) $\{y : y \geq 0\}$ (d) $\{y : y < 0\}$

51. A graph that meets every horizontal line in at most one point is the graph of a function that is

(a) one-to-one (b) onto (c) constant (d) even

53. The inverse of $f(x) = \frac{1}{x}$ for $x \neq 0$ is

(a) x (b) $-\frac{1}{x}$ (c) x^2 (d) $\frac{1}{x}$ (that is, f is its own inverse)

55. If $f(x) = 2x + 5$, then $f^{-1}(5) =$

(a) 5 (b) 2 (c) 0 (d) 15

57. If f has an inverse, then $(f \circ f^{-1})(4) =$

(a) 2 (b) 4 (c) 16 (d) 0

59. If $f(x) = x^2 - 4$, then $f(3) =$

(a) 5 (b) 9 (c) 13 (d) 12

(a) $f(x) = 2x$ (b) $f(x) = \frac{1}{x}$ (c) $f(x) = x^2$ (d)
 $f(x) = x + 1$

60. An equivalence relation is reflexive,
symmetric and

(a) antisymmetric (b) transitive (c) surjective (d)
complete